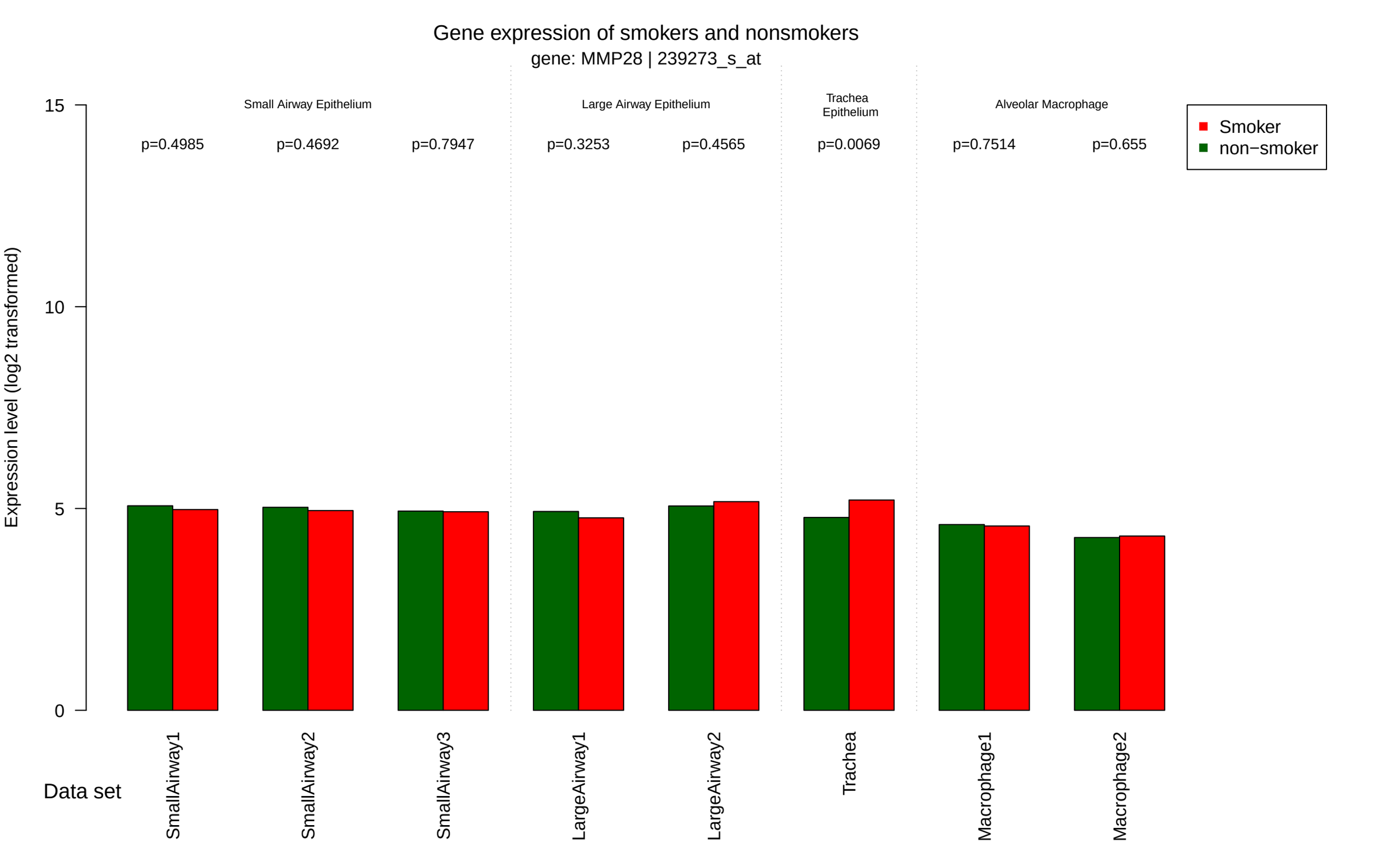
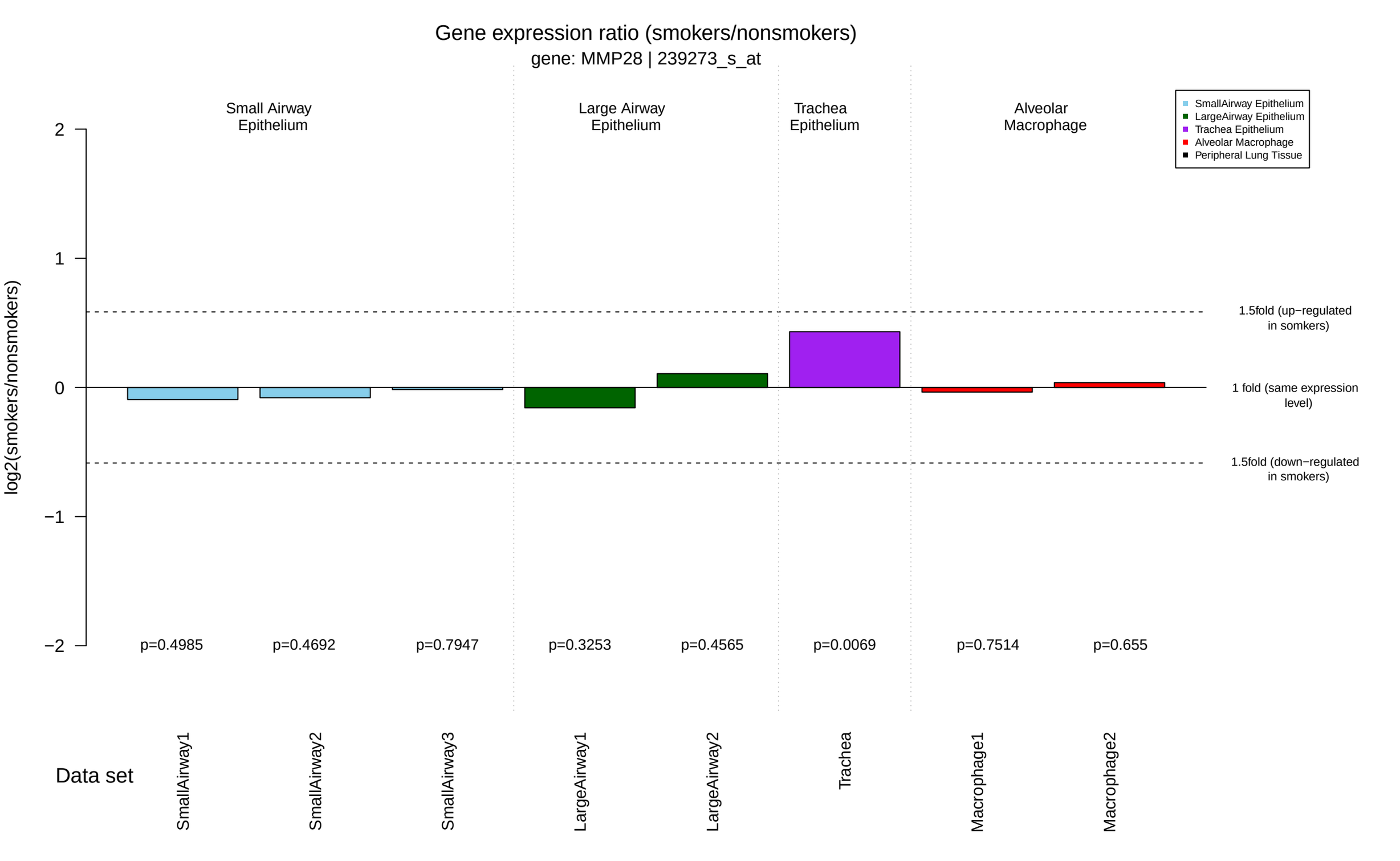
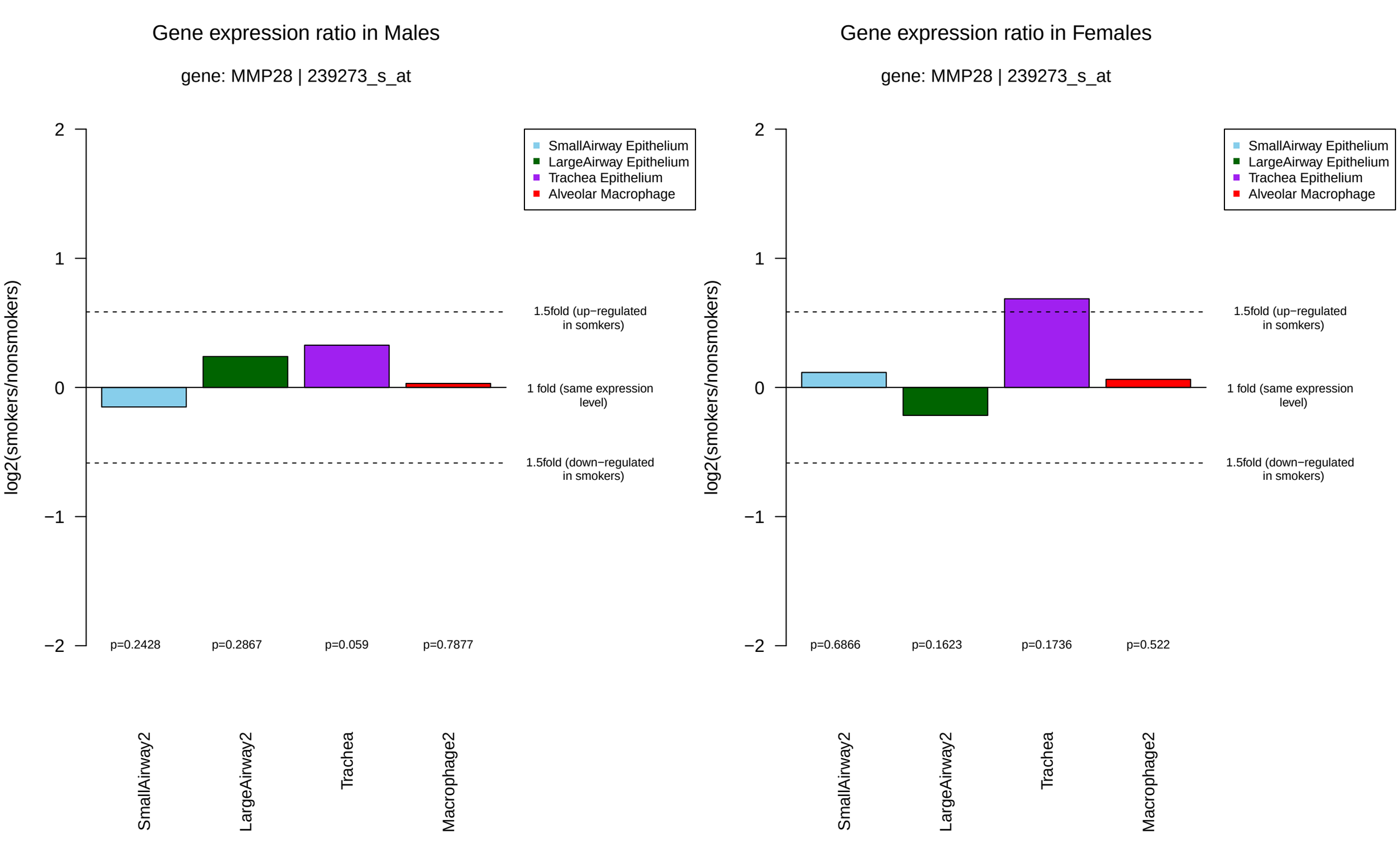
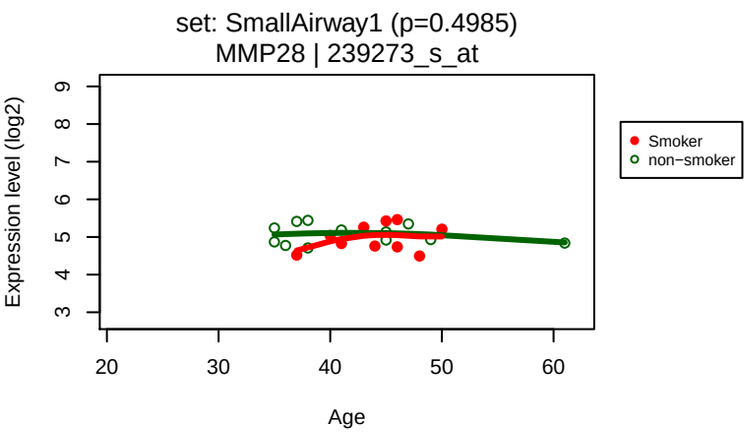
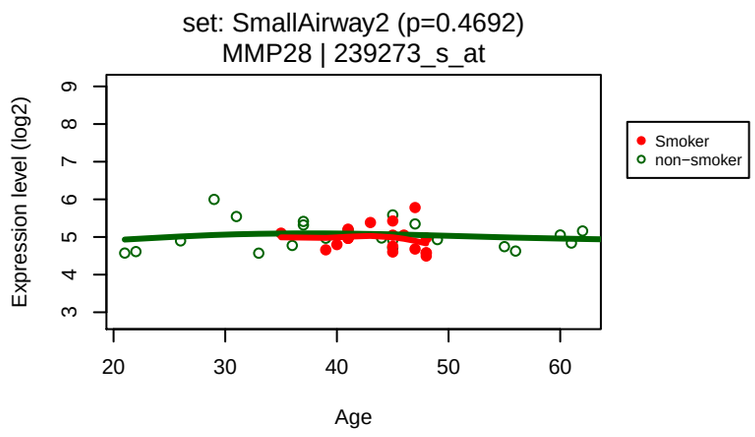
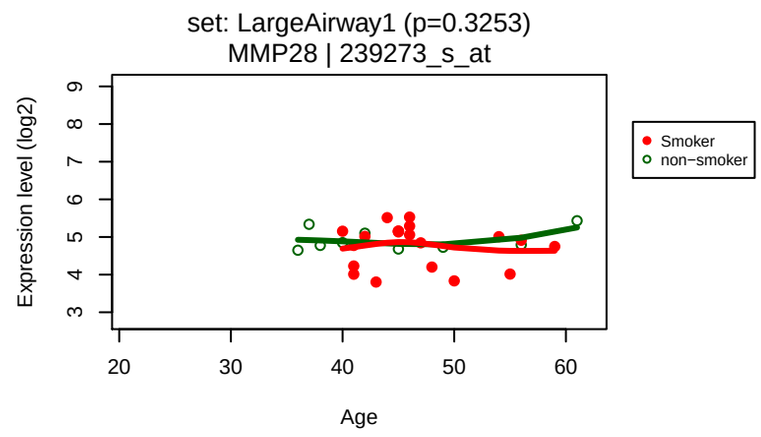

Gene expression of smokers and nonsmokers
gene: MMP28 | 239273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4985
p=0.4692
p=0.7947
p=0.3253
p=0.4565
p=0.0069
p=0.7514
p=0.655
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

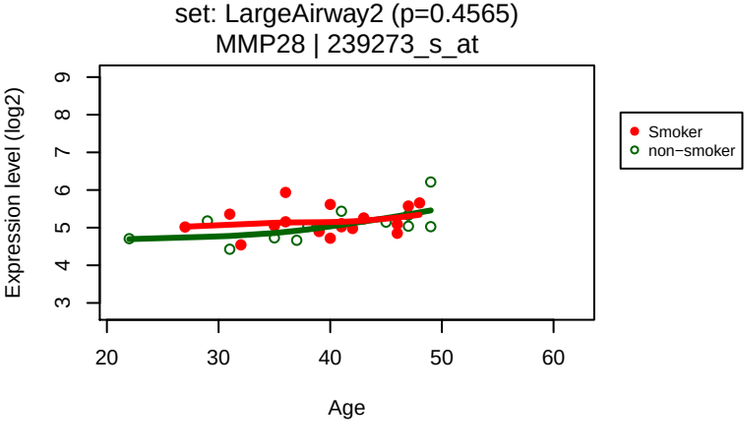

Gene expression ratio (smokers/nonsmokers)
gene: MMP28 | 239273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4985
p=0.4692
p=0.7947
p=0.3253
p=0.4565
p=0.0069
p=0.7514
p=0.655
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

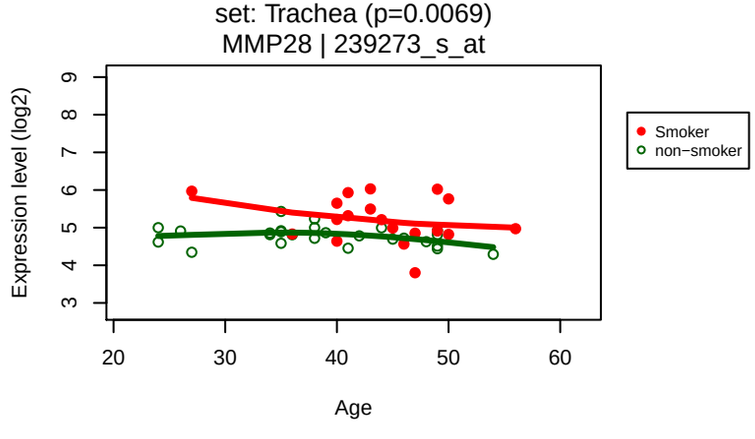

Gene expression ratio in Males
gene: MMP28 | 239273_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2428
p=0.2867
p=0.059
p=0.7877
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP28 | 239273_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6866
p=0.1623
p=0.1736
p=0.522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

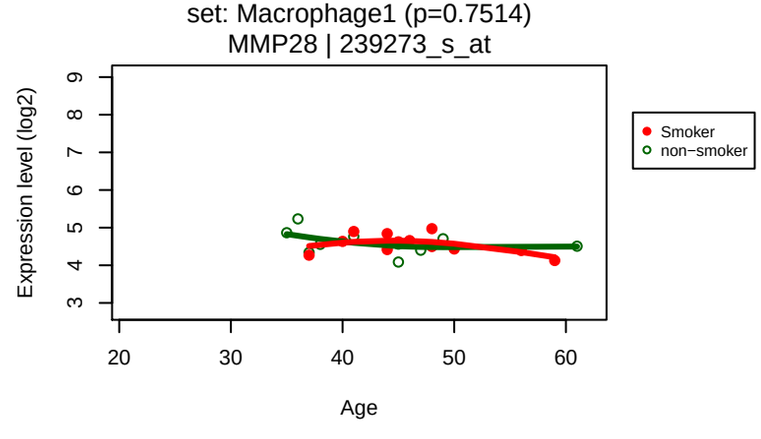

set: SmallAirway1 (p=0.4985)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker

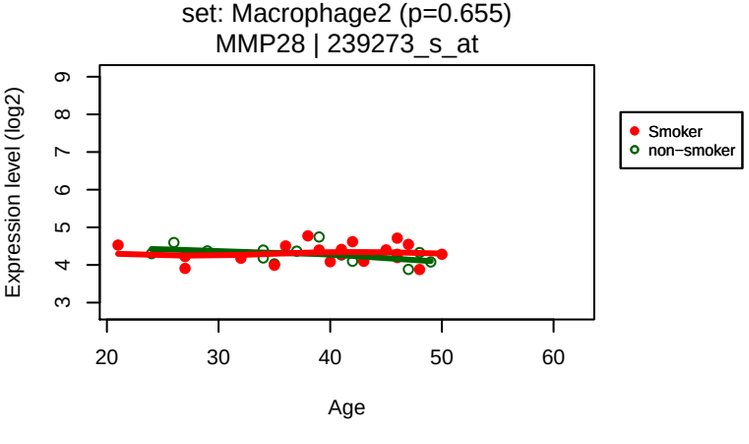

set: SmallAirway2 (p=0.4692)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker

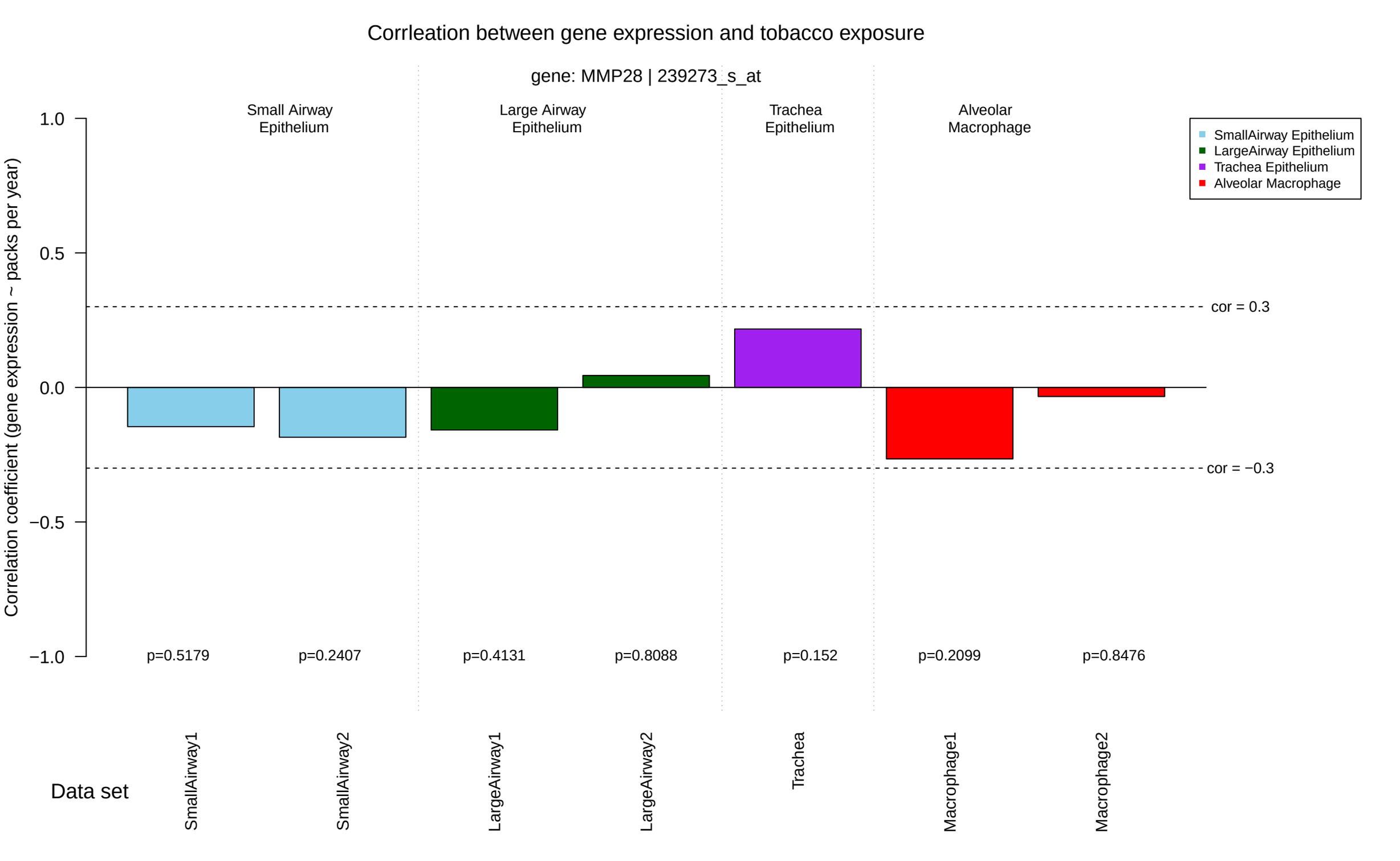

set: LargeAirway1 (p=0.3253)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4565)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0069)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7514)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.655)
MMP28 | 239273_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP28 | 239273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5179
p=0.2407
p=0.4131
p=0.8088
p=0.152
p=0.2099
p=0.8476
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2